| | | | | |
|---|---|---|---|---|
| OR-AS Operations Research Applications and Solutions | Case Name: **Railway Station (2)** | Sector | **Construction (civil)** | |
| | **OR-AS**<br>Operations Research - Applications and Solutions<br>www.or-as.be<br>info@or-as.be | Baseline Schedule | Schedule with resources | |
| | | | Schedule with costs | |
| | | Risk Analysis | Random simulation | |
| | | | One of nine std. scenarios | |
| Submitted by | **Matilde Casado and Margarida Antunes** | | User defined distributions | |
| Date | **2015** | Project Control | Automatic tracking | |
| File Name | **C2015-09 Railway Station (2)** | | Tracking based on user input | |

# 1. Project description

Project authenticity

A railway station construction project involving foundation, infrastructure, superstructure, and finishing works for modernization and accessibility.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

# 2. Project properties

## *2.1. Baseline Schedule*

| General | |
|---|---|
| # Activities | 348 |
| Planned Duration (PD) | 354 days* |
| Budget At Completion (BAC) | 1 457 424.0€ |
| Renewable Resources | - |
| Consumable Resources | - |

* standard eight-hour working days

| Network topology | |
|---|---|
| Serial/Parallel (SP) | 4% |
| Activity Distribution (AD) | 48% |
| Length of Arcs (LA) | 0% |
| Topological Float (TF) | 75% |

## *2.2. Risk Analysis*

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

| | Cost sensitivity | | |
|---|---|---|---|
| | avg [%] | std dev [%] | skew [-] |
| CRI-r | 5.8 | 8.3 | -2.7 |
| CRI-rho | 25.0 | 20.9 | -0.3 |
| CRI-tau | 92.0 | 22.7 | -0.5 |

| | Resource sensitivity | | |
|---|---|---|---|
| | avg [%] | std dev [%] | skew [-] |
| CRI-r | N/A | N/A | N/A |
| CRI-rho | N/A | N/A | N/A |
| CRI-tau | N/A | N/A | N/A |

| | Time sensitivity | | |
|---|---|---|---|
| | avg [%] | std dev [%] | skew [-] |
| CI | 4.3 | 13.7 | -3.3 |
| SI | 13.6 | 21.5 | -2.1 |
| SSI | 1.3 | 4.7 | -5.3 |
| CRI-r | 8.6 | 6.9 | -1.2 |
| CRI-rho | 8.7 | 6.6 | -1.3 |
| CRI-tau | 8.2 | 6.3 | -0.9 |

## *2.3. Project Control*

### *2.3.1. Simulated forecasting accuracy*

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

| **Simulated EAC(t) accuracy** | | |
|---|---|---|
| method - PF | MAPE [%] | MPE [%] |
| PV - 1 | 28.8 | 28.1 |
| PV - SPI | 49.3 | 48.8 |
| PV - SCI | 49.8 | 48.8 |
| ED - 1 | 52.7 | 51.8 |
| ED - SPI | 49.3 | 48.7 |
| ED - SCI | 49.4 | 48.8 |
| ES - 1 | 23.8 | 23.5 |
| ES - SPI(t) | 36.5 | 36.3 |
| ES - SCI(t) | 36.5 | 36.3 |

| **Simulated EAC accuracy** | | |
|---|---|---|
| method (PF) | MAPE [%] | MPE [%] |
| 1 | 2.3 | 0.0 |
| CPI | 2.5 | 0.3 |
| SPI | 19.5 | 19.4 |
| SPI(t) | 17.5 | 17.4 |
| SCI | 19.5 | 19.5 |
| SCI(t) | 17.6 | 17.5 |
| 0.8 CPI + 0.2 SPI | 9.0 | 8.9 |
| 0.8 CPI + 0.2 SPI(t) | 9.1 | 8.6 |

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

### *2.3.2. Tracking description*

Tracking authenticity

Manual tracking was performed over 9 tracking periods with a length of approximately 3 months. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

## *2.3.3. Earned Value Management*

### 2.3.3.1. Performance metrics

| | CV [€] | SV [€] | SV(t) [d] | CPI [-] | SPI [-] | SPI(t) [-] | p-factor [-] |
|---|---|---|---|---|---|---|---|
| avg | -194018.9 | -385965.0 | -1013.0 | 0.8 | 0.6 | 0.6 | 0.9 |
| std dev | 222839.3 | 383228.2 | 515.1 | 0.1 | 0.3 | 0.1 | 0.1 |
| final | -688253.0 | -403182.1 | -1728.0 | 0.7 | 1.0 | 0.6 | 1.0 |

### 2.3.3.2. Time forecasting

| PD | 354 days | Real Duration | 570 days | Late | 61.0% |
|---|---|---|---|---|---|

| EAC(t) | | | Real Accuracy | |
|---|---|---|---|---|
| method - PF | avg [d] | std dev [d] | MAPE [%] | MPE [%] |
| PV - 1 | 362.2 | 2.1 | 4.1 | 4.1 |
| PV - SPI | 362.7 | 2.1 | 4.0 | 4.0 |
| PV - SCI | 363.4 | 2.1 | 4.0 | 4.0 |
| ED - 1 | 364.2 | 1.9 | 4.0 | 4.0 |
| ED - SPI | 365.0 | 1.9 | 4.0 | 4.0 |
| ED - SCI | 365.7 | 1.9 | 4.0 | 4.0 |
| ES - 1 | 366.5 | 2.1 | 4.0 | 4.0 |
| ES - SPI(t) | 367.2 | 2.1 | 4.0 | 4.0 |
| ES - SCI(t) | 367.7 | 2.1 | 3.9 | 3.9 |

### 2.3.3.3. Cost forecasting

| BAC | 14 574 234.0€ | Real Cost | 2 145 682.3€ | Over Budget | 47.2% |
|---|---|---|---|---|---|

| EAC | | | Real Accuracy | |
|---|---|---|---|---|
| method (PF) | avg [€] | std dev [€] | MAPE [%] | MPE [%] |
| 1 | 1651448.2 | 222839.3 | 2.6 | 2.6 |
| CPI | 1862549.4 | 182035.0 | 1.5 | 1.5 |
| SPI | 2646603.8 | 1051066.2 | 2.6 | -2.6 |
| SPI(t) | 2389528.9 | 278309.9 | 1.3 | -1.3 |
| SCI | 3121589.1 | 1413224.6 | 5.1 | -5.1 |
| SCI(t) | 2771365.5 | 475311.3 | 3.2 | -3.2 |
| 0.8 CPI + 0.2 SPI | 1939169.5 | 190188.6 | 1.1 | 1.1 |
| 0.8 CPI + 0.2 SPI(t) | 1936197.7 | 149353.9 | 1.1 | 1.1 |